

Week 7 – Day 1

Task 4: Fish Audio Evaluation

1. Introduction

This evaluation compares Fish Audio and ElevenLabs for a production-grade AI voice agent designed for a Pakistani real estate company. The evaluation focuses on conversational quality, real-time performance, multilingual capabilities, and suitability for natural UrduLish interactions.

2. Evaluation Criteria

Criterion	Fish Audio	ElevenLabs	Preferred
Latency	Excellent real-time streaming performance.	Excellent, especially with real-time/Flash models.	Tie
Naturalness	Highly natural conversational voices.	Excellent, particularly for English narration.	Slight Edge: ElevenLabs
Emotion	Expressive emotional speech with natural intonation.	Excellent emotional delivery.	Tie
Streaming	Designed for low-latency conversational streaming.	Supports real-time streaming.	Tie
Voice Cloning	High-quality, fast voice cloning.	Industry-leading voice cloning.	Slight Edge: ElevenLabs
Pricing	Competitive pricing for production workloads.	Generally higher cost for similar usage.	Fish Audio
Multilingual Support	Strong multilingual speech generation.	Excellent multilingual support.	Tie
Urdu Pronunciation	Very suitable for Urdu speech.	Good, but more English-centric.	Fish Audio
Urdu-English Switching	Smooth UrduLish code-switching.	Good but less natural in mixed-language conversations.	Fish Audio

3. Fish Audio Strengths

- Expressive and emotionally natural speech.
- Strong multilingual capabilities with smooth Urdu-English code-switching.
- Excellent streaming support for low-latency voice agents.
- High-quality voice cloning.
- Competitive pricing for production deployments.
- Natural pauses and conversational intonation.

4. ElevenLabs Strengths

- Outstanding English voice quality.
- Industry-leading voice cloning.
- Large voice library and mature ecosystem.
- Strong emotional speech generation.

5. Recommended Voice Stack

- Speech-to-Text: Deepgram Nova-3 (Whisper Large V3 for offline scenarios)
- LLM: GPT-5.5, Claude, or Gemini
- Agent Framework: LangGraph
- Knowledge Layer: ChromaDB or Pinecone with RAG
- Text-to-Speech: Fish Audio
- Workflow Automation: n8n
- Calendar: Google Calendar API
- Email: Gmail API or Microsoft Graph API
- Backend: FastAPI
- Deployment: Docker + Railway/Render/AWS

6. Recommendation

Based on the project requirements and the evaluation above, Fish Audio is the recommended text-to-speech engine. Its expressive conversational voices, strong multilingual capabilities, smooth Urdu-English code-switching, competitive pricing, and low-latency streaming make it particularly well suited for a Pakistani real estate voice agent. Although ElevenLabs remains an excellent platform, especially for English narration and premium voice cloning, Fish Audio better aligns with the project's primary objective of delivering natural UrduLish conversations.

7. Conclusion

Fish Audio offers the best balance of conversational quality, multilingual performance, streaming capability, and affordability for this capstone. Therefore, it is selected as the preferred Text-to-Speech solution.